Name	Michel Fadel
Contact Number	michelfadel04@gmail.com
Project Title (Example – Week1, Week2, Week3)	Week 2: Raw Sales Data

Project Guidelines and Rules

1. **Formatting and Submission**
   - **Format:** Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
   - **Title:** Include Week and Title (Example - Week 1: TravelEase Case Study.)
   - **File Format:** Submit as PDF or Word file to contact@victoriasolutions.co.uk
   - **Page Limit:** 4–5 pages, including the title and references.

2. **Answer Requirements**
   - **Word Count:** Each answer should be 100–150 words; total 800–1,200 words.
   - **Clarity:** Write concise, structured answers with key points.
   - **Tone:** Use formal, professional language.

3. **Content Rules**
   - Answer all questions thoroughly, referencing case study concepts.
   - Use examples where possible (e.g., risk assessment techniques).
   - Break complex answers into bullet points or lists.

4. **Plagiarism Policy**
   - Submit original work; no copy-pasting.
   - Cite external material in a consistent format (e.g., APA, MLA).

5. **Evaluation Criteria**
   - **Understanding:** Clear grasp of business analysis principles.
   - **Application:** Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
   - **Clarity:** Logical, well-structured responses.
   - **Creativity:** Innovative problem-solving and examples.
   - **Completeness:** Answer all questions within the word limit.

6. **Deadlines and Late Submissions**
   - **Deadline:** Submit on time; trainees who submit fail to submit the project will miss the "Certificate of Excellence"

7. **Additional Resources**
   - Refer to lecture notes and recommended readings.
   - Contact the instructor or peers for clarifications before the deadline.

START YOUR PROJECT FROM HERE:

1) Cleaned Dataset:

1. Dropped the duplicate record (John Doe).

2. Filled the Nan values in the Email and Phone columns with 'Unknown' because they are personal data, so they can't be left null or filled with random values.

3. Filled the Nan values in the Discount column with the mean.

4. Changed the format of the Order_Date column to (dd/mm/yy).

5. Renamed the Discount (%) column to Percentage_Discount.

6. Separated the Order_Date column into three columns 'Day', 'Month' and 'Year' for better analysis.

Order_ID	Customer_Name	Email	Phone	Product_Category	Day	Month	Year	Revenue	Percentage_Discount
101	John Doe	john@email.com	9876543210	Electronics	31	12	2023	1200	10
102	Alice Smith	Unknown	9898989898	Clothing	5	1	2024	500	15
103	Bob Miller	bob@email.com	Unknown	Electronics	12	1	2024	3000	20
105	David White	david@email.com	9123456789	Furniture	15	2	2024	2500	15
106	Emma Brown	emma@email.com	9234567890	Clothing	8	3	2024	700	5
107	Chris Green	Unknown	9345678901	Furniture	10	4	2024	1800	25
108	Alice Smith	alice@email.com	Unknown	Clothing	8	3	2024	500	15

2) Exploratory Data Analysis Summary:

1. The product category with the highest revenue is Furniture while the lowest is Clothing.

```
SELECT Product_Category, SUM(Revenue) AS 'Total_Revenue' FROM sales_data
GROUP BY Product_Category
ORDER BY SUM(Revenue) DESC
;
```

2. The Furniture category has the highest average discounts and Clothing has the lowest average discounts.

```
SELECT Product_Category, AVG(Percentage_Discount) AS 'Avg_Discount' FROM sales_data
GROUP BY Product_Category
ORDER BY AVG(Percentage_Discount) DESC
;
```

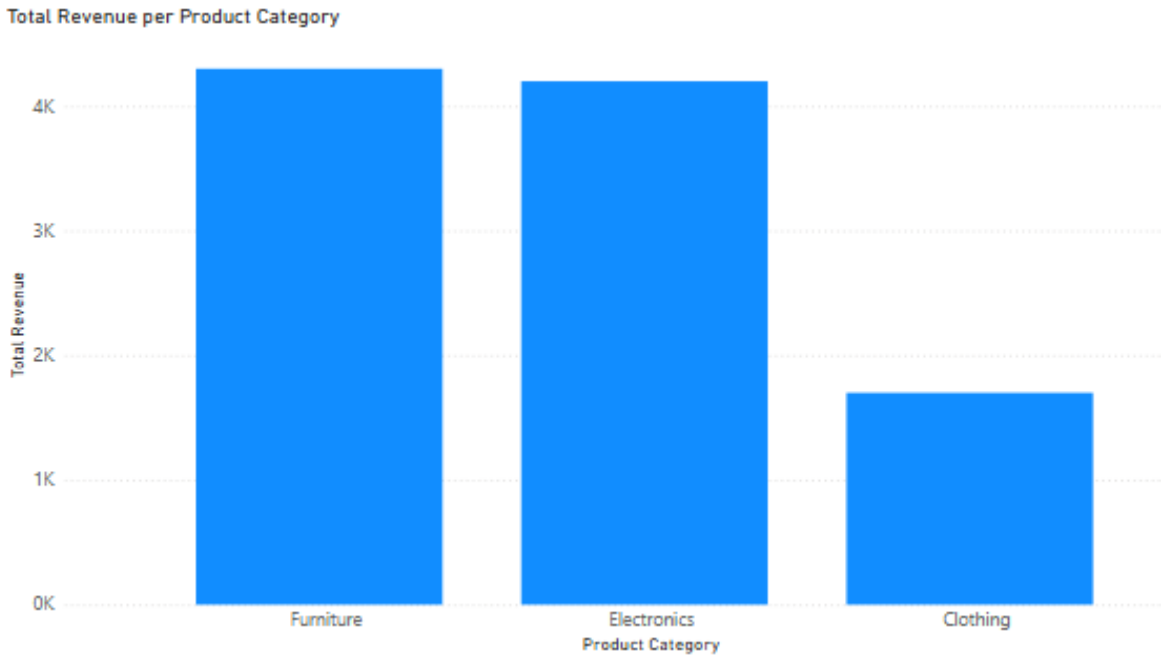
3. The average revenue per month increases gradually from December to February (having the highest revenue/month) then decreases the following month and rises again.

```
SELECT AVG(Revenue) AS 'Avg_Revenue',  Month FROM sales_data
GROUP BY Month
ORDER BY AVG(Revenue)
;
```

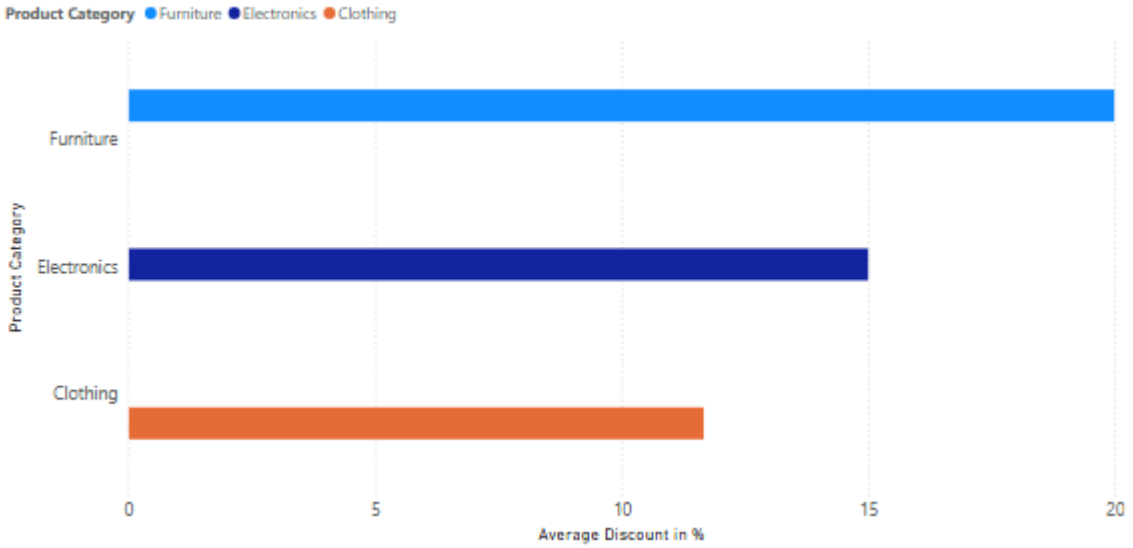
4. The values change constantly between the Revenue and the Discount columns as the revenue increases, so there is no correlation between those two columns.

```
SELECT Revenue, Percentage_Discount FROM sales_data
GROUP BY Revenue, Percentage_Discount
ORDER BY Revenue
;
```

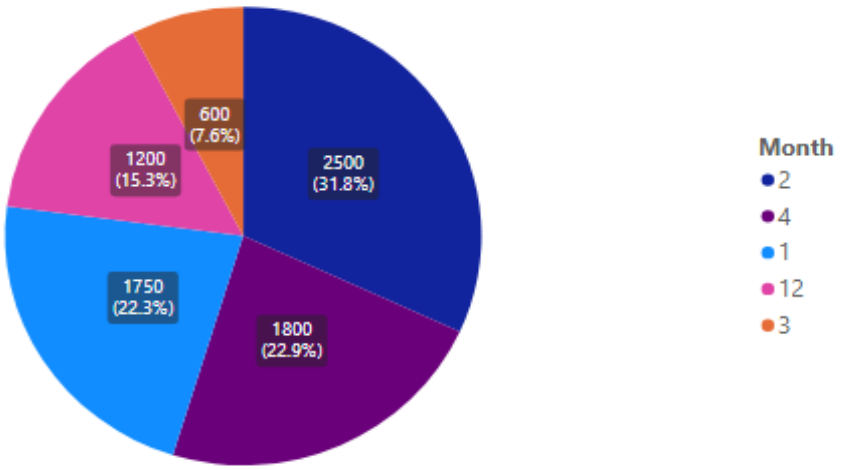
3) Data Visualizations:

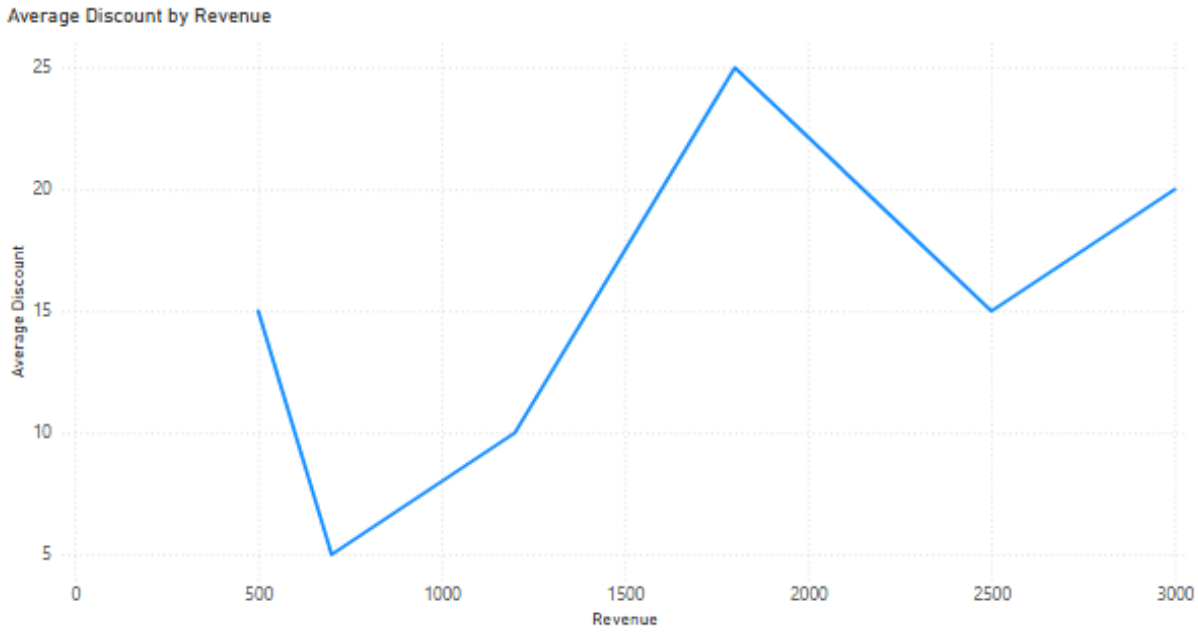


Average Discount Applied on Each Product Category



Average Revenue per Month





4) Data Insights Summary:

The total Revenue of all sales is £10,200 while the average discount applied on all products is 15%. The category with the highest revenue is Furniture (£4,300) possibly because it had the highest average discount of 20% followed by the Electronics with a revenue of £4,200 and an average discount of 15% and lastly Clothing with the lowest revenue of £1,700 and an average discount of nearly 12%.

The average revenue increases gradually from December to February (which is the peak month) due to holiday season then drops in March likely due to previous purchasing period and rises again in April. The relationship between the revenue and the average discount is not steady so as the revenue increases the average discount sometimes increases and sometimes decreases, therefore there is no correlation between the two variables.

Recommendations can be provided such as:

- Increase marketing campaigns on the Clothing category especially from December to February to increase sales.
- Increase stock for electronic and furniture products as they are the most sold categories.
- Apply better discounts and offers in March to increase sales.